

texture that absorbs all the minute details of the input and allows for even the most discrete of elements to be identified in the output.

Going back to the image case, the same image will be obtained at the output in some possible configurations, albeit with slight differences from the actual input so that they can be used in realistic examples. Users may be familiar with image captchas where the entire image is split up into equally sized fragments and requires certain elements to be identified. It's one way of understanding how they work.

Convolutional neural networks are similar to feedforward neural networks with all layers having learnable weights and biases. The neural network features are taken batch-wise like water passing through a filter.

This allows the system to remember the images in smaller parts and can compute the operations. Again, these conversions are of large order as the images are converted from colour (RGB or HIS) to grey-scale. Once this is executed, any alterations in the pixel value will alert the system of any classification changes.

WHY ARE CONVOLUTIONAL NETWORKS SO GREAT

While industry experts would argue about the stalemate disadvantages that come with CNNs, they can't deny that they are so useful. Before moving on to the good, let's explore some reasons why CNNs get a bad rap in the industry:-

High Computational Cost:- CNNs don't exactly work with numbers and strings but heavier data that raises the computational needs for a system by a huge margin. It's no secret that software development costs for image detection apps are considerably larger to simpler apps. Research has shown that the accuracy of deep learning models increases with larger datasets, which in turn increase the computational and memory requirements. Many frameworks today optimize CNNs to the underlying hardware to achieve maximum efficiency. For e.g.: The training could be performed on general purpose x86 platforms built using the Intel® Xeon® Scalable processor family and then inference is performed on more power efficient hardware such as the Intel® Movidius™ NCS.

Hardware Requirements:- There are different hardware platforms in the market today that can be used for training and inference. One could look at standard x86 hardware platforms, GPUs, TPUs, FPGAs, accelerators such as Intel® Movidius™ VPUs, and others. The choice depends on final deployment scenarios that will need to be considered.